

LATHURZAN SUBATHARAN

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Summary

First Class BSc (Hons) Software Development graduate of Cardiff Metropolitan University with hands-on experience in full-stack development, AI integration, and production deployment. Built and shipped an award-winning AI-powered travel platform selected for the CST Expo 2026 (top 15 of 200+ submissions), validated by 253 automated tests. Seeking an internship, graduate, or junior software engineering role to contribute immediately across backend, full-stack, and AI-integrated systems.

Education

Cardiff Metropolitan University

Cardiff, Wales, UK

BSc (Hons) Software Development — **First Class Honours (1st)**

2023 – 2026

- Dissertation project (Place Finder AI) selected for CST Expo 2026 — top 15 of 200+ submissions university-wide
- Key modules: Data Structures & Algorithms, Software Architecture, Database Systems, Agile Development

ICBT Campus

Sri Lanka

Higher National Diploma (HND) in Software Engineering

2025

Experience

Full-Stack Developer (Contract)

Jan 2025 – May 2025

HNV Construction

Sri Lanka

- Built a full-stack task and project management platform using **React** and **Node.js**, replacing manual spreadsheet workflows for the operations team
- Designed and implemented **RESTful APIs** for real-time data sync between frontend and backend, reducing data inconsistency errors across the system
- Deployed the application to a **Linux VPS** production environment, achieving stable uptime and enabling remote team access across office locations
- Automated backend business workflows, cutting manual data processing effort by an estimated **35–40%** across recurring reporting tasks

Projects

Place Finder AI — *React · TypeScript · FastAPI · PostgreSQL · PostGIS · Gemini AI · scikit-learn · Stripe*

2026

- ★ **CST Expo 2026** — **Top 15 of 200+ projects** at Cardiff Metropolitan University, recognised for AI innovation and full-stack engineering
- GitHub: github.com/Lathurzan/place-finder-ai
- Engineered an AI travel discovery platform integrating **Google Gemini AI** for natural language search, voice input, conversational assistance, itinerary generation, and image-based destination recognition
- Built a scalable **async FastAPI** backend with JWT authentication, image upload processing, and non-blocking PostgreSQL operations handling concurrent requests without latency degradation
- Implemented a personalised **recommendation engine** using TF-IDF vectorisation and cosine similarity (scikit-learn), delivering context-aware suggestions with measurably higher relevance than keyword search
- Developed **geospatial proximity search** using PostgreSQL 15 + PostGIS with ST_DWithin and GIST indexing, returning results in **<100ms** across datasets exceeding 10,000 records
- Authored **129 pytest** unit/integration tests and **124 Postman** API tests across auth, AI services, geocoding, weather, and payment routes — full route coverage

Community Health & Services Portal (CHaSP) — *React · Spring Boot · Flask · MySQL · OpenAI API*

2025–2026

- GitHub: github.com/Lathurzan/Community-Health-Services-Portal
- Developed a multi-tier healthcare platform spanning React frontend, Spring Boot REST API, and Flask microservice — supporting patient, staff, and admin roles
- Implemented **JWT authentication** and **RBAC** securing all protected routes across three distinct user types
- Integrated **OpenAI API** chatbot for automated user support; **Stripe** and **PayPal** for secure online transactions

Book Shop Management System — *Java · JSP · Servlets · MySQL*

2025

- GitHub: github.com/Lathurzan/ap
- Built a full MVC Java web application with authentication, inventory management, billing, and transaction history — demonstrating solid OOP design and relational database integration

Technical Skills

Languages	Python, Java, JavaScript, TypeScript, SQL
Frontend	React 18, TypeScript, Tailwind CSS, Leaflet.js, HTML5, CSS3
Backend	FastAPI (async), Spring Boot, Node.js, Flask, REST API Design, JWT, RBAC
Databases	PostgreSQL, PostGIS, MySQL, MongoDB, GIST Indexing, Spatial Queries
AI / ML	Google Gemini AI, OpenAI API, TF-IDF Vectorisation, Cosine Similarity, scikit-learn, Prompt Engineering
Cloud & DevOps	AWS EC2, AWS S3, VPS Deployment, Linux, CI/CD Fundamentals, Git, GitHub
Testing	Pytest (129 tests), Postman (124 tests), JUnit
Concepts	OOP, MVC, Microservices, Async Programming, Agile/Scrum, System Design, DSA

Achievements

CST Expo 2026 — Top 15 of 200+ Projects (Cardiff Metropolitan University)	2026
Place Finder AI recognised for innovation in AI-powered discovery, ML recommendations, and geospatial engineering	
National Gold Medalist — Martial Arts	Sri Lanka
National-level gold medal demonstrating discipline, resilience, and high performance under pressure	